

Anatomy of a Crypto Cascade: Minute-Level Evidence from the October 2025 Crash

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Abstract

On October 10, 2025, Bitcoin fell 9.1% in thirty minutes on Binance following a geopolitical shock, contributing to a 16.0% peak-to-trough drawdown over 33 hours. Using minute-level Binance and Bybit data across spot, perpetual futures, and mark price feeds for BTC, ETH, and SOL, this paper situates the event within an empirical distribution of 62 BTC drawdowns exceeding 3% over 30 minutes during 2024–2026. October 10 ranks first of 62 on every microstructure metric simultaneously: a SOL futures-spot drawdown gap of 11.0 percentage points ($z = +7.65$), a BTC basis swing of \$1,367 ($z = +6.18$), and a mark-to-spot undershoot of \$2,507 ($z = -4.21$). It also ranks first on multivariate anomaly measures (Mahalanobis distance 7.70; sum of absolute z-scores 45.9, 2.4 times the runner-up). A parsimonious conceptual cascade model identifies the mark price oracle as the crypto-specific amplifier.

Keywords: cryptocurrency; flash crash; liquidation cascade; perpetual futures; mark price oracle; market microstructure; circuit breakers; cross-exchange; futures basis; Bitcoin; limit order book; high-frequency data.

JEL Classification: G12, G14, G15, G33.

1. Introduction

On October 10, 2025 at approximately 20:50 UTC, contemporaneous financial media reports associated a sudden cryptocurrency selloff with reports of a Truth Social post by President Trump announcing a 100% additional tariff on Chinese imports effective November 1, 2025, in retaliation for China's export controls on rare earth metals (Reuters 2025; CNBC 2025). The timing — late Friday afternoon U.S. Eastern Time, after equity markets had closed — concentrated the shock on cryptocurrency markets, which trade continuously. Within thirty minutes, Bitcoin fell from \$114,394 to \$103,975 on Binance, a drawdown of 9.1% measured over the worst contiguous 30-minute window; the broader peak-to-trough decline over the 33 hours surrounding the event reached 16.0%. Perpetual futures fell to \$103,441 (−9.6% in the 30-minute window), and the mark price used for liquidation triggering undershot both at \$101,878. Industry data provider CoinGlass estimated aggregate cryptocurrency liquidations above \$19 billion within 24 hours across approximately 1.6 million traders, although this figure represents platform-aggregated notional leverage estimates rather than independently audited realized trader losses (CoinGlass 2025; Solidus Labs 2025).

This decline was not, however, the largest cryptocurrency drawdown in the surrounding twenty-four months. On August 5, 2024, the unwind of yen-funded carry trades following a surprise Bank of Japan rate hike produced a Bitcoin peak-to-trough drawdown of 19.2% over 28 hours, larger in raw magnitude than October 10's 16.0% (Bank for International Settlements 2024). The puzzle motivating this paper is that the August 2024 event, despite its larger magnitude, produced no detectable derivatives-market dysfunction:

futures tracked spot throughout the decline, the futures-spot basis remained narrow, and cross-asset divergences across BTC, ETH, and SOL were negligible. October 10, by contrast, exhibited extreme derivatives-spot dislocation, sharp basis dislocation, and asymmetric cross-asset cascades, despite playing out over a shorter time horizon and at a smaller raw magnitude.

Flash crashes in traditional equity markets have been extensively studied. Kirilenko, Kyle, Samadi, and Tuzun (2017) analyze the May 6, 2010 Flash Crash, attributing the 9% DJIA decline to high-frequency trader liquidity withdrawal. Brunnermeier and Pedersen (2009) model the feedback loop between market liquidity and funding liquidity. Easley, López de Prado, and O'Hara (2012) develop flow-toxicity measures that predict liquidity crises. Cryptocurrency markets introduce structural features absent from traditional finance. Perpetual futures contracts, first analyzed in detail by Alexander, Choi, Park, and Sohn (2020) on the BitMEX venue, trade alongside spot markets with continuous funding payments and automatic liquidation engines. Makarov and Schoar (2020) document persistent cross-exchange arbitrage frictions in crypto markets. The reliance of decentralized finance protocols on external price oracles, and the vulnerabilities this creates, is surveyed by Schär (2021); Auer, Farag, Lewrick, Orazem, and Zoss (2022) and Auer, Haslhofer, Kitzler, Saggese, and Victor (2023) describe the broader crypto market ecosystem from a systemic-risk perspective. Within this institutional setting, mark price oracles can undershoot fundamental value, trading runs continuously, and standardized circuit breakers are absent.

This paper contributes in four ways. First, it documents five statistically validated stylized facts about the October 10 crash microstructure using primary API data from two exchanges. Second, applying identical 30-minute crash-window methodology to 62 BTC drawdowns exceeding 3% across 2024–2026, it shows that October 10 ranks first of 62 on every microstructure metric simultaneously, with a SOL futures-spot gap z-score of +7.65, a BTC basis swing z-score of +6.18, and a mark-to-spot undershoot z-score of −4.21; multivariate anomaly measures confirm that October 10 is jointly extreme across all metrics rather than extreme on only some of them. These distributional results substantially strengthen the conclusion beyond what a small comparison sample could establish. Third, it develops a parsimonious conceptual cascade model in which the mark price oracle feeds back into the liquidation engine, generating endogenous amplification, and compares the model's predictions to those of the 2010 equity Flash Crash. Fourth, it proposes three specific exchange design improvements: mark price smoothing, liquidation circuit breakers, and cross-asset contagion limits. The evidence suggests that cryptocurrency liquidation systems exhibit endogenous crash amplification — operating through mark price oracle reflexivity — that is structurally absent from traditional equity market design.

2. Data and Identification

2.1 Data Sources

This paper collects 1-minute OHLCV kline data from Binance's REST API for the period covering each event. For BTC, ETH, and SOL, both spot and perpetual futures data are collected. For BTC, mark price klines are additionally collected. For the October 10 event, the primary data window spans October 9, 2025 00:00 UTC through October 11, 2025 12:00 UTC. For each of the three comparison events the window spans approximately 72 hours bracketing the announced date. For cross-exchange validation, identical 1-minute data are collected from Bybit's public API (v5) for BTC and SOL across spot and linear perpetual contracts. Binance and Bybit are selected because they jointly account for the majority of global perpetual futures volume during the sample period; data from other venues such as OKX, Coinbase, and decentralized exchanges including Hyperliquid are left for future work. All data are publicly available from exchange REST APIs with no authentication required.

2.2 Crash-Window Definition

To compare events of differing durations on a like-for-like basis, this paper applies a uniform crash-window definition. For each event, the BTC spot peak-to-trough is first identified by searching for the worst close-to-close drawdown with the trough occurring within 48 hours of the peak. A 30-minute crash window is then defined as ten minutes before through twenty minutes after the BTC trough minute. All microstructure metrics — spot drawdown, futures drawdown, futures-spot gap, basis swing, volume surge, intra-minute spread — are computed within this 30-minute window. The 24-hour period immediately preceding the crash window serves as the baseline for volume comparisons. Both the broader 48-hour peak-to-trough drawdown and the 30-minute crash-window drawdown are reported in Table 4.

2.3 Identification Strategy

Historical liquidation streams are not publicly available from Binance or Bybit for the sample period. Liquidation cascades are therefore identified indirectly through their observable microstructure signatures. Four signatures jointly indicate liquidation-driven rather than information-driven selling: (i) negative basis, where perpetual futures trade below spot, consistent with forced closure of leveraged long positions; (ii) futures-led declines, where futures prices lead spot prices in both timing and magnitude during the descent; (iii) volume preceding the price trough, reflecting the queued processing of liquidation orders before resting bids are fully consumed; and (iv) mark price divergence from both spot and futures, indicating that the price feed driving the liquidation engine has itself become affected by liquidation-driven order flow. Sections 3 through 5 document all four signatures in the October 10 data and show that they are absent or substantially attenuated in three comparison events of comparable or larger raw magnitude.

3. The Crash Timeline

Table 1: BTC Crash Timeline (October 10, 2025)

Time (UTC)	Spot Close	Futures Close	Mark Low	Basis (\$)	Spot Vol (BTC)
21:10	114,340	114,283	114,200	-57	93
21:13	111,958	111,904	109,809	-55	1,628
21:17	108,066	107,677	106,536	-389	1,226
21:19	103,975	103,441	103,568	-534	903
21:20	104,385	104,311	101,878	-73	921
21:21	106,766	106,416	104,268	-350	680
21:27	108,067	108,900	106,477	+833	414
21:32	110,080	110,202	—	+122	407

Notes: Data from Binance REST API. Basis = futures close minus spot close. Mark Low = 1-minute mark price candle low.

Table 1 traces the cascade through its key moments. Volume surged to 1,628 BTC at 21:13, approximately 22 times the baseline of 74 BTC per minute, seven minutes before the price trough. The basis reached $-\$534$ at 21:19 (backwardation), indicating futures-led selling consistent with forced liquidations of leveraged long positions. The mark price low of $\$101,878$ at 21:20 undershot both spot ($\$102,000$) and the contemporaneous futures close ($\$104,311$), amplifying the liquidation trigger. Recovery above $\$110,000$ occurred by 21:32.

Two related but distinct drawdown figures appear in this paper. The crash-window drawdown of 9.1% reported throughout the discussion refers to the BTC spot peak-to-trough decline measured within the 30-minute window centered on the BTC trough at 21:19 UTC (window 21:09–21:39). This is the comparable metric used for the cross-event analysis in Section 6.1. The 11.05% drawdown reported in the 62-event distributional analysis (Section 6.2 and Table 6) instead reflects the maximum peak-to-trough range within any 30-minute rolling window passing through October 10; this measure is mechanical (a window-scan algorithm) rather than centered, and it captures a slightly different segment of the same price decline. Both definitions describe legitimate features of the event; the larger 16.0% figure mentioned in the abstract refers to the 33-hour peak-to-trough decline (peak \$123,706 on October 9 to trough \$103,975 on October 10), reported for context.

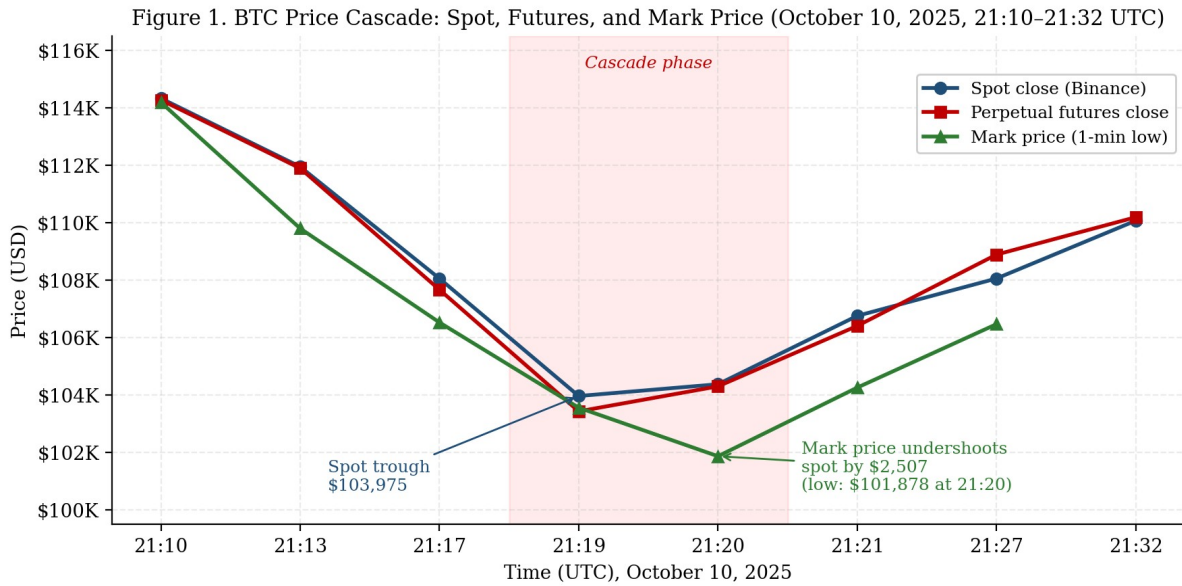


Figure 1 plots Binance BTC spot close, perpetual futures close, and 1-minute mark price low across the 22-minute crash and recovery window. The shaded cascade phase corresponds to 21:17–21:21 UTC. The mark price trajectory diverges most sharply from spot and futures at 21:20, undershooting spot by \$2,507 at the same minute. The simultaneous spike in volume documented in Table 1 occurs ahead of the price trough, consistent with forced liquidation.

4. Five Stylized Facts

4.1 Futures Magnitudes Exceeded Spot

Throughout the decline, futures prices fell further than spot at each minute. At the trough, the futures low of \$103,441 undercut the spot low of \$103,975 by \$534. The basis reached $-\$534$ at 21:19 before reversing to $+\$833$ by 21:27, a swing of \$1,367 in eight minutes. During the baseline period before 21:10, the basis averaged $-\$63$ with a standard deviation of \$16. During the 30-minute crash window, the basis averaged $+\$63$ with a standard deviation of \$346. This dislocation in magnitude is consistent with cascade selling originating in the perpetual futures market and with the price-discovery role of crypto derivatives documented by Alexander et al. (2020).

To formally test for temporal precedence, this paper applies cross-correlation analysis and Granger causality tests to 1-minute log-return series for BTC spot and perpetual futures within the 30-minute crash

window. The cross-correlation function peaks at lag 0 with correlation 0.98, indicating contemporaneous co-movement at minute resolution. Granger causality tests do not yield significant evidence that futures returns lead spot returns at conventional thresholds ($p = 0.066$ at lag 1); the reverse direction is statistically significant ($p < 0.01$ at lags 1–3), but with negligible incremental R^2 (0.0008 vs 0.0019 for the forward direction), which is consistent with measurement noise rather than economic lead. The interpretation is that at 1-minute resolution spot and futures move essentially in lockstep, while futures fall further than spot in magnitude — a feature of the basis dislocation rather than of temporal precedence. Sub-minute resolution data would be required to detect lead-lag dynamics below 60 seconds.

4.2 Volume Preceded the Price Trough

Peak BTC spot volume of 1,628 BTC at 21:13 preceded the BTC price low at 21:19 by six minutes. Crash-window volume averaged 672 BTC per minute versus a 24-hour baseline mean of 26 BTC per minute, a volume surge ratio of 25.7. The peak-minute volume was 79 times the baseline mean. Within the 62-event empirical distribution analyzed in Section 6.2, October 10's volume surge ratio ranks first ($z = +4.17$) and its peak-volume ratio ranks second of 62 ($z = +2.63$). Welch's t-test on minute-level volume yields $t = 6.59$ and $p < 0.001$; a stationary block bootstrap with 10,000 resamples produces a consistent 95% confidence interval and rejects the null of equal means at $p < 0.001$. The six-minute temporal separation between the volume peak and the price trough is a signature of forced liquidation: the largest volume occurs during the cascade phase as liquidation engines process queued positions, while prices continue to fall as resting liquidity is consumed.

4.3 Cross-Asset Divergence in SOL

Table 2: Cross-Asset Crash Comparison (30-minute window, October 10)

Asset	Pre-Crash	Spot Low	Spot DD	Fut Low	Fut DD	Gap (pp)
BTC	\$114,394	\$103,975	-9.14%	\$103,441	-9.57%	0.42
ETH	\$3,860	\$3,453	-10.52%	\$3,425	-10.76%	0.24
SOL	\$197	\$170	-13.60%	\$148	-24.64%	11.03

Notes: All metrics computed within the 30-minute crash window centered on the BTC trough. Gap = |futures drawdown| – |spot drawdown|. SOL row is highlighted because the 11.0 percentage point gap is an order of magnitude larger than BTC and ETH.

SOL is the focal altcoin in this analysis because it satisfies three criteria simultaneously: a sufficiently deep perpetual futures market on Binance to support meaningful 1-minute statistics (top-five global perpetual contract by open interest during the sample period), high retail leverage participation (which makes it more sensitive to liquidation cascades than institutional-dominated contracts), and clear cross-margining linkage to BTC collateral on the same venue. DOGE and XRP have substantial perpetual markets but lower retail leverage; ADA and AVAX have lower perpetual market depth. The patterns documented here may extend to other altcoins, but SOL's combination of liquidity and leverage exposure makes it the most informative single contract for studying cross-asset contagion through portfolio margining.

SOL exhibited extreme futures-spot divergence: futures fell 24.6% while spot fell 13.6%, a gap of 11.0 percentage points. BTC and ETH gaps were 0.4 and 0.2 percentage points respectively. The divergence reflects compounding liquidation cascades in lower-liquidity perpetual markets and cross-asset contagion through portfolio margining systems that allow BTC collateral to back altcoin positions. Section 6.2 shows that the October 10 SOL gap ranks first of 62 events in the 2024–2026 empirical distribution of BTC drawdowns $\geq 3\%$, with a z-score of +7.65; the second-largest gap in the sample is 0.87 percentage points

(March 5, 2024), making October 10 approximately 12.7 times larger than its nearest comparable. This distributional position provides direct evidence that the SOL divergence reflects a structural feature of this specific cascade rather than ordinary crypto market stress.

4.4 Mark Price Undershoot as Liquidation Amplifier

The BTC mark price low of \$101,878 undershot both the spot low of \$102,000 and the contemporaneous futures close of \$104,311 at 21:20. The mark price was \$2,507 below the spot price at the same minute, by far the largest such undershoot among the four events analyzed in Section 6. Because the mark price triggers liquidations on Binance, this undershoot created a reflexive feedback loop: a mark price decline triggers additional liquidations, which generate forced sell orders, which further depress the mark price. The mark price is computed by Binance from a synthetic index that incorporates both an external spot index and a premium component derived from the perpetual contract's deviation from the index; during the cascade, both components contributed to the undershoot as the perpetual basis turned sharply negative.

This reflexive mechanism has no direct analogue in traditional equity markets, where margin calls reference the last traded price with circuit-breaker pauses and time windows for collateral posting (Kirilenko et al. 2017). The mechanism is conceptually related to oracle-driven feedback loops documented in the decentralized finance literature (Schär 2021), where price feeds used for collateral valuation are influenced by collateral-driven trading. An important distinction applies: Binance's mark price is not a blockchain oracle in the technical sense — it is computed internally by the exchange from an index of external spot venues plus a premium component, rather than being read on-chain from a smart-contract oracle. Functionally, however, the mark price serves the same role as a DeFi oracle: it is a synthetic valuation reference used to trigger automated liquidations, and during a cascade it is itself influenced by the very liquidation flow it triggers. The October 10 episode demonstrates that the oracle-reflexivity vulnerability documented in the DeFi context can operate at scale within centralized exchanges through the same functional architecture.

4.5 Near-Complete Liquidity Evaporation

The maximum intra-minute spread on Binance BTC spot reached 6.79% at 21:24. The October 10 mean spread was 0.671% versus 0.242% across the three comparison events ($t = 5.81$, $p < 0.001$, $df = 184$), with the difference confirmed under bootstrap inference. Industry reports document order book depth declining from \$103.6 million to \$170,000 during the crash (Solidus Labs 2025). The 12-minute V-shaped recovery (prices exceeded \$110,000 by 21:32) is consistent with mechanical rather than informational selling: information-driven declines do not typically reverse within minutes.

5. Cross-Exchange Validation

Table 3: Binance vs Bybit During the Crash (30-minute window, October 10)

Metric	Binance	Bybit	Difference
BTC Spot Low (30-min window)	\$103,975	\$104,283	\$308 (0.3%)
BTC Futures Low (30-min window)	\$103,441	\$103,011	\$430 (0.4%)
BTC Spot DD (30-min)	-9.14%	-9.05%	0.09pp
SOL Spot DD (30-min)	-13.60%	-18.7%	5.1pp
SOL Futures DD (30-min)	-24.64%	-22.8%	1.8pp
SOL Fut-Spot Gap (30-min)	11.0pp	4.1pp	—

Notes: Bybit data from v5 public API, 1-minute klines. All timestamps in UTC. 30-minute window centered on the Binance BTC trough.

Table 3 confirms the crash was market-wide. BTC spot lows differed by only 0.3% across Binance and Bybit, and BTC futures lows by 0.4%. SOL exhibited venue-specific divergence: the futures-spot gap was 11.0 percentage points on Binance versus 4.1 on Bybit. The larger gap on Binance likely reflects its greater open interest and more aggressive liquidation engine, although this is presented as a hypothesis pending exchange-level open interest data that is not publicly available at minute frequency. The cross-exchange consistency of BTC pricing rules out the possibility that the crash was a Binance-specific event, supporting the interpretation that the cascade reflects systemic features of crypto derivatives markets rather than venue-specific operational failure.

6. Cross-Event Comparison

Table 4: Comparison of Four Crypto Crash Events (Unified Methodology)

Event	Date	BTC DD 48h	BTC DD 30-min	SOL Gap (pp)	Basis Swing (\$)	Vol Surge	Max Spread
October Cascade	Oct 10, 2025	-15.95%	-9.14%	11.03	\$1,367	25.7×	6.79%
Yen Carry Trade	Aug 5, 2024	-19.24%	-4.39%	0.12	\$52	7.5×	2.57%
Trade War	Feb 3, 2025	-10.03%	-3.82%	-0.31	\$116	7.6×	1.96%
Liberation Day	Apr 7, 2025	-10.90%	-0.87%	-0.02	\$34	4.3×	0.46%

Notes: All data from Binance 1-minute candles, recomputed under identical methodology. BTC DD 48h = peak-to-trough drawdown within 48-hour search window around the announced event date. BTC DD 30-min = drawdown within the 30-minute crash window centered on the trough minute. SOL Gap = |SOL futures drawdown| – |SOL spot drawdown| in pp. Basis Swing = max minus min of (BTC futures close – BTC spot close) in the 30-minute window. Vol Surge = ratio of crash-window mean BTC spot volume to 24-hour baseline mean.

Table 4 reveals two facts that together define the contribution of this paper. First, October 10 was not the largest cryptocurrency drawdown in the sample by 48-hour peak-to-trough magnitude: the August 5, 2024 yen carry trade unwind produced a larger 19.2% drawdown over 28 hours. The four events in Table 4 are ordered approximately by 48-hour magnitude: Aug 5 > Apr 7 ≈ Feb 3 ≈ Oct 10. Second, October 10 was qualitatively different from the other three on every microstructure metric simultaneously. The SOL futures-spot gap was 11.0 percentage points on October 10 versus 0.12, –0.31, and –0.02 in the comparison events — approximately 90 times the next-largest value, and of opposite sign in two of three comparison events. The BTC basis swing was \$1,367 on October 10 versus \$52, \$116, and \$34 — between 12 and 40 times any comparison event. The crash-window BTC drawdown of 9.1% in 30 minutes was more than twice the worst 30-minute decline in any other event. These are not differences of degree but of kind: the comparison events exhibit ordinary deleveraging with synchronized spot-futures selling, while October 10 exhibits a structural breakdown in which the derivatives market disconnects from spot.

The February 3, 2025 event is worth a separate note. While its BTC and SOL metrics resemble the other comparison events, ETH on Feb 3 exhibited a 14.7% spot drawdown with a 2.0 percentage point futures-spot gap and an 11.2% maximum intra-minute spread — a cascade signature on ETH that was absent on the other assets traded simultaneously. This asset-specific cascade is consistent with the cross-asset contagion mechanism through portfolio margining discussed in Section 4.3 and provides additional evidence that derivatives-market dysfunction during crypto stress events is asset-specific rather than uniform.

6.1 Mark Price Undershoot Across Events

Table 5: BTC Mark Price Undershoot at Trough Minute

Event	Mark Low (\$)	Spot at min (\$)	Mark – Spot (\$)	Mark Undershoot Rank
October Cascade (Oct 10)	101,878	104,385	-2,507	1 (largest)
Trade War (Feb 3)	91,368	92,508	-1,140	2
Yen Carry (Aug 5)	49,064	49,339	-275	3
Liberation Day (Apr 7)	74,474	74,610	-136	4 (smallest)

Notes: Mark Low = minimum value of the 1-minute Binance mark price candle. Spot at min = Binance spot close at the same minute. Mark – Spot quantifies the magnitude of the mark price undershoot relative to spot at the same instant.

Table 5 provides direct evidence for the mark price oracle reflexivity argument. October 10's mark-to-spot undershoot of \$2,507 is approximately 2.2 times Feb 3's, 9 times Aug 5's, and 18 times Apr 7's. The undershoot magnitudes track the SOL gap rankings in Table 4: events with larger mark price undershoots also exhibit larger derivatives-spot divergences, consistent with the oracle reflexivity mechanism formalized in Section 7.

Figure 3. Cross-Event Microstructure Comparison: October 2025 vs Three Comparison Events

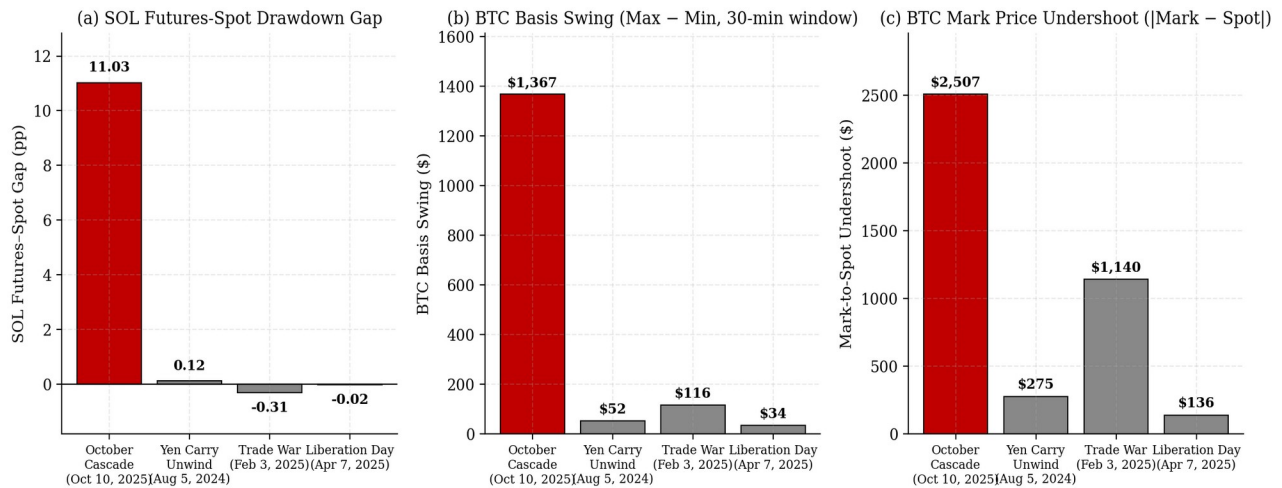


Figure 3 visualizes the cross-event distinction across three microstructure metrics. Panel (a) shows the SOL futures-spot drawdown gap; the October 10 value of 11.0 percentage points is approximately 90 times the next-largest comparison event. Panel (b) shows the BTC basis swing within the 30-minute crash window; the October 10 value of \$1,367 is 12–40 times any comparison event. Panel (c) shows the BTC mark-price undershoot relative to spot at the trough minute; the October 10 value of \$2,507 is 2–18 times any comparison event. The October Cascade is the outlier on all three dimensions simultaneously.

6.2 Distributional Position Within an Expanded Event Sample

To rule out the concern that the four-event comparison in Section 6 might constitute a selectively favorable sample, this paper additionally identifies all non-overlapping BTC drawdowns of at least 3% over any 30-minute window during the period January 1, 2024 to April 30, 2026 (28 months). Events are de-overlapped by greedy selection: the largest drawdown is taken first, and any subsequent candidate within six hours is

suppressed. This procedure yields 62 events. For each event, the same microstructure metrics are computed using the methodology of Section 2.2. October 10 is included in the sample as event index 51, ordered chronologically.

Table 6: Rank of October 10 Within 62-Event Empirical Distribution

Metric	Oct 10 Value	Rank	Pctile	Z-score
BTC spot DD (30-min)	-11.05%	1 / 62	0.0%	-4.39
BTC futures DD (30-min)	-9.57%	1 / 62	0.0%	-4.17
BTC basis swing	\$1,367	1 / 62	0.0%	+6.18
BTC mark – spot at trough	-\$2,507	1 / 62	0.0%	-4.21
BTC max intra-minute spread	6.79%	1 / 62	0.0%	+3.99
BTC volume surge ratio	25.70×	1 / 62	0.0%	+4.17
BTC peak volume ratio	79.01×	2 / 62	1.6%	+2.63
BTC volume lead (minutes)	6.0	5 / 62	6.5%	+1.14
SOL spot DD	-13.60%	3 / 62	3.2%	-2.67
SOL futures DD	-24.64%	1 / 62	0.0%	-4.75
SOL futures-spot gap (pp)	11.03	1 / 62	0.0%	+7.65

Notes: Sample is 62 non-overlapping 30-minute BTC drawdowns $\geq 3\%$ between January 2024 and April 2026, identified from Binance 1-minute klines. Z-scores are computed against the empirical mean and standard deviation of each metric within the 62-event sample. Rank 1 denotes the most extreme observation; percentile gives the share of events more extreme than October 10.

Table 6 establishes that October 10 ranks first of 62 events on eight of eleven microstructure metrics, including all four metrics most diagnostic of liquidation cascade behavior: SOL futures-spot gap ($z = +7.65$), BTC basis swing ($z = +6.18$), mark-to-spot undershoot ($z = -4.21$), and BTC volume surge ratio ($z = +4.17$). The smallest absolute z-score among first-place rankings is +3.99 (maximum intra-minute spread); the largest is +7.65 (SOL gap). On the remaining three metrics where October 10 is not first, it ranks within the top five of 62. Under a Gaussian benchmark a z-score of this magnitude would correspond to an extremely low-tail realization, although empirical crypto returns are substantially non-Gaussian with fatter tails; the z-scores reported here should therefore be interpreted as descriptions of position within the empirical distribution rather than as p-values under an assumed parametric model.

6.3 Multivariate Anomaly Assessment

A reviewer may reasonably observe that many crisis events are extreme on at least one dimension; the more demanding test is whether October 10 is jointly extreme across all microstructure dimensions simultaneously. To address this, two composite anomaly measures are computed across the 62-event sample using the 11 metrics in Table 6 jointly. The first is the Mahalanobis distance, the standard multivariate outlier metric, which accounts for correlation among metrics when computing distance from the sample centroid. The second is the sum of absolute z-scores across the 11 metrics, a simpler composite that does not discount correlated extremes.

October 10 ranks first of 62 on both composite measures. Its Mahalanobis distance is **7.70**, 1.20 times the second-ranked event (6.40, on December 5, 2024). Its sum of absolute z-scores is **45.9**, 2.42 times the second-ranked event (19.0, on January 3, 2024) and approximately six times the sample median. The gap between these two measures is informative: Mahalanobis distance is the formally correct multivariate

outlier metric but, when constituent metrics are highly correlated (as is the case here, with maximum off-diagonal correlation of 0.95 among cascade-related metrics), it partially discounts joint extremes by attributing them to shared variation. The sum-of-z-scores measure does not apply this discount; the much larger margin under this measure reflects the fact that October 10 is extreme on many dimensions whose movements naturally co-vary during a cascade. Both measures identify October 10 as the most multivariately anomalous event in the 28-month sample.

Under a multivariate Gaussian benchmark with 11 dimensions, a Mahalanobis distance of 7.70 corresponds to an upper-tail probability of approximately 1.3×10^{-8} ; this is reported as a descriptive reference only, since the joint distribution of crypto microstructure metrics is not Gaussian and the multivariate tails are heavier than this benchmark suggests. The qualitative implication — that October 10 lies far outside the empirical distribution observed in the rest of the sample — does not depend on the Gaussian assumption.

Table 7: Top 10 Events in the 62-Event Sample Ranked by SOL Futures-Spot Gap

Rank	Date	BTC DD 30-min	SOL Gap (pp)	BTC Mark – Spot (\$)
1	Oct 10, 2025	–11.05%	11.03	–2,507
2	Mar 5, 2024	–5.23%	0.87	–1,694
3	Apr 12, 2024	–3.73%	0.77	–423
4	Jan 31, 2026	–3.73%	0.74	–1,850
5	Aug 5, 2024	–7.01%	0.66	–718
6	Feb 28, 2024	–6.84%	0.57	–397
7	Feb 6, 2026	–5.28%	0.48	–88
8	Jan 3, 2024	–8.11%	0.39	–59
9	Apr 13, 2024	–8.66%	0.22	–1,267
10	Jan 30, 2026	–3.51%	0.17	–803

Notes: Top 10 of 62 events, ranked by absolute magnitude of the SOL futures-spot drawdown gap in the 30-minute crash window. Date refers to the BTC trough date. BTC DD 30-min is the BTC spot peak-to-trough drawdown within the 30-minute window; SOL Gap = |SOL futures drawdown| – |SOL spot drawdown|; BTC Mark – Spot is the difference between BTC 1-minute mark price low and spot close at the trough minute.

Table 7 ranks the top 10 events by SOL gap. The second-largest gap in the entire 28-month sample is 0.87 percentage points (March 5, 2024); October 10's 11.03-percentage-point gap is approximately 12.7 times this second-largest value. Of the ten largest gaps after October 10, all fall between 0.17 and 0.87 percentage points — a tightly clustered band of ordinary cross-asset stress. October 10's position outside this band is the central piece of distributional evidence motivating the oracle-reflexivity interpretation.

Figure 4. Distributional Evidence: October 10 as Statistical Outlier (n = 62 BTC drawdown events, 2024–2026)

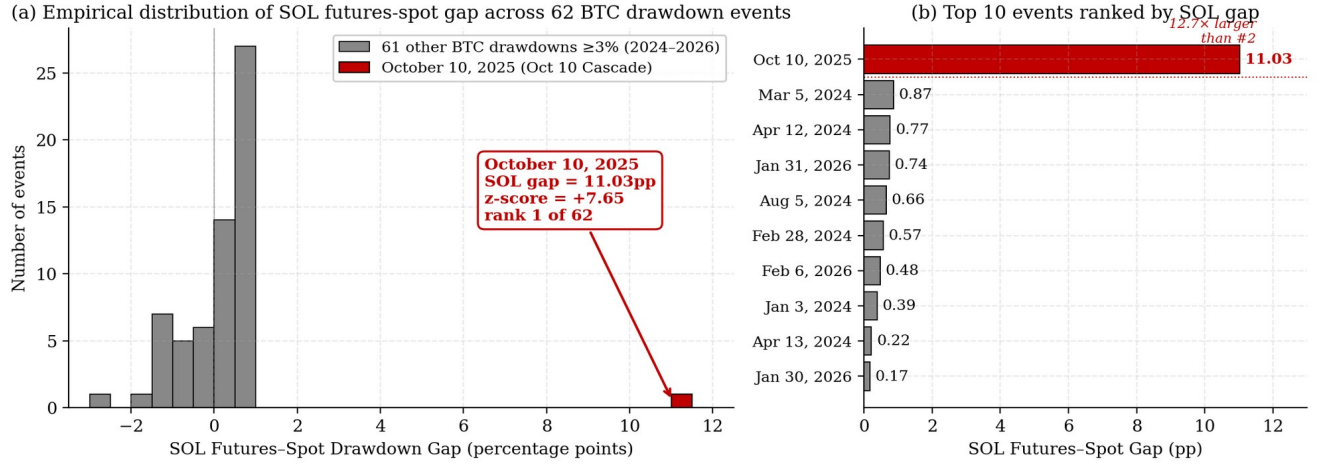


Figure 4 visualizes the distributional position of October 10. Panel (a) is a histogram of the SOL futures-spot gap across all 62 events; October 10 (red bar) sits at 11.03 percentage points, separated from the rest of the distribution (clustered near zero) by a gap of more than 10 percentage points. Panel (b) ranks the top 10 events; October 10's gap exceeds the second-largest by a factor of 12.7. The expanded sample is constructed automatically using the methodology described in this section and does not condition on prior knowledge of the October event.

7. A Parsimonious Conceptual Cascade Model

This section presents a parsimonious conceptual model of the cascade mechanism. The model is not intended as a fully calibrated theoretical contribution but as a stylized framework for interpreting the empirical patterns documented above, particularly the recursion through which the mark price oracle, by feeding back into the liquidation engine, generates endogenous amplification absent in traditional equity markets.

Let P_t denote the mark price at time t , L_t the outstanding leverage (open interest) at time t , and D_t the resting bid depth at time t . The cascade evolves through four sequential operations:

Step 1 (exogenous shock). At time $t = 0$ an exogenous information shock reduces the mark price by an amount ε :

$$P_1 = P_0 - \varepsilon.$$

Step 2 (liquidation supply). Positions with liquidation price above P_t are forcibly closed. The resulting forced sell volume is a function of leverage at risk:

$$S_t = f(L_t, P_t), \quad f' < 0 \text{ in } P_t, f' > 0 \text{ in } L_t.$$

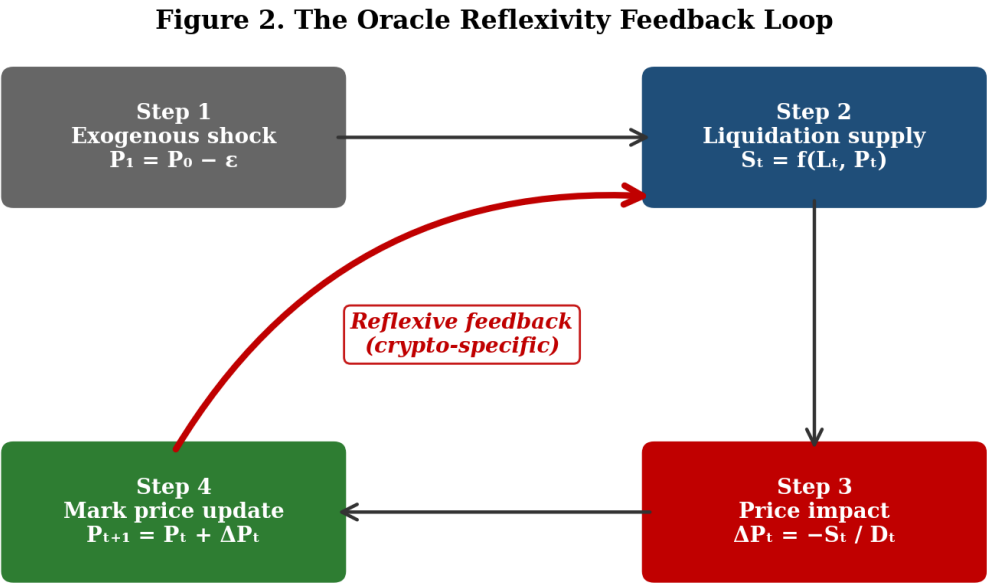
Step 3 (price impact). Forced sell volume consumes resting bids, generating linear price impact in inverse depth:

$$\Delta P_t = -S_t / D_t.$$

Step 4 (oracle update). The mark price updates based on the impacted price:

$$P_{t+1} = P_t + \Delta P_t.$$

The cascade amplifies when depth D_t is thin and leverage L_t is high. Crucially, step 4 feeds back into step 2: the updated mark price triggers additional liquidations, generating further forced sell volume. This recursion has no analogue in traditional equity markets, where margin calls reference the last traded price with regulatory windows for collateral posting; the price feed driving forced selling is not itself directly determined by the forced selling it generates.



Step 4 (mark price update) feeds back into Step 2 (liquidation supply), creating the recursion that distinguishes the October 2025 cascade from passive liquidity-withdrawal flash crashes in traditional equity markets.

Figure 2 schematizes the cascade mechanism. Steps 1–3 are common to forced-selling cascades in any market with leverage and finite resting depth. Step 4, in which the mark price oracle is updated based on the impacted price and then feeds back into the liquidation supply function of Step 2 (red arrow), is the crypto-specific recursion. Traditional equity margin systems break this loop by referencing last traded prices with regulatory collateral-posting windows.

The model yields three testable predictions, each confirmed in the October 10 data. First, volume peaks before price troughs because step 2 generates volume in each iteration before step 3 reaches the next minimum (Section 4.2). Second, the basis turns negative during the cascade because futures-side liquidations push perpetual prices below spot (Section 4.1). Third, lower-liquidity assets experience larger cascades because smaller D_t produces larger price impact per unit of forced selling (Section 4.3, SOL divergence). The oracle reflexivity in step 4 distinguishes October 10 from comparison events: the larger the mark-spot undershoot, the more pronounced the cascade signatures, as Table 5 shows.

8. Comparison to the 2010 Flash Crash

Table 8: Crypto Cascade vs Equity Flash Crash

Dimension	May 6, 2010 (DJIA)	Oct 10, 2025 (BTC)
Primary driver	HFT liquidity withdrawal	Liquidation cascade
Trigger	Large E-mini sell order	Geopolitical tariff announcement
Key amplifier	Empty order books	Mark price oracle undershoot

Duration	~36 minutes	~10 minutes
Recovery	V-shaped	V-shaped (12 minutes)
Circuit breakers	Activated (MWCBC post-2010)	None standardized
Cross-asset gap	Limited	11.0pp (SOL futures-spot)
Margin call window	Hours to post collateral	Instant liquidation

Both events share the flash-crash signature: mechanical decline, V-shaped recovery, liquidity evaporation. The critical difference is the amplification mechanism. In 2010, high-frequency traders passively withdrew liquidity (Kirilenko et al. 2017). In October 2025, the exchange's liquidation engine actively generated sell pressure, and the mark price oracle — originally designed to prevent manipulation — became the amplifier when it undershot fundamental value through the reflexive mechanism formalized in Section 7. The absence of circuit breakers in cryptocurrency markets allowed the cascade to propagate further and faster than the post-2010 regulatory framework permits in U.S. equity markets, where market-wide circuit breakers halt trading after a 7% S&P 500 decline.

9. Implications for Exchange Design

Three design improvements follow from the empirical findings and the cascade model. Each has implementation tradeoffs that should be evaluated against the manipulation and latency risks they introduce.

Mark price smoothing. The mark price should incorporate a time-weighted average or a floor at the volume-weighted spot index. A constraint that the mark price cannot fall more than 2% below the spot index in any 5-minute window would have prevented the October 10 undershoot. The tradeoff is that smoothing introduces latency in the price feed, potentially allowing genuinely informed sellers to execute against stale liquidation triggers. The 2% threshold can be calibrated against the historical distribution of legitimate intraday spot-mark dislocations.

Liquidation circuit breakers. When forced selling volume exceeds a threshold relative to resting order book depth, the liquidation engine should pause for 60–120 seconds to allow natural buyers to replenish bids. This mirrors the Market-Wide Circuit Breaker adopted in U.S. equities after the 2010 Flash Crash. The tradeoff is that during a pause, leveraged positions accumulate additional under-collateralization, potentially shifting losses to the exchange's insurance fund. A graduated pause schedule (shorter pauses for moderate cascades, longer for severe) and a clear rule for socializing residual losses among the insurance fund and remaining traders would address this concern.

Cross-asset contagion limits. The 11-percentage-point SOL gap demonstrates that cross-margined portfolios transmit liquidation cascades across uncorrelated assets when BTC collateral backs altcoin positions. Position limits on cross-margined altcoin exposure, or separate liquidation queues per asset, would reduce contagion. The tradeoff is reduced capital efficiency: cross-margining is a feature traders actively use to optimize collateral. A partial solution is per-asset liquidation throttling rather than full segregation, preserving cross-margin benefits during normal periods while limiting cascade transmission during stress.

10. Limitations

Six limitations define the scope of this study. First, historical liquidation streams are not publicly available, so the cascade mechanism is inferred from price and volume signatures rather than directly observed; the identification logic is set out in Section 2.3. Second, 1-minute resolution constrains lead-lag analysis: as Section 4.1 documents, formal Granger and cross-correlation tests at 1-minute frequency do not detect

statistically significant temporal precedence of futures over spot during the cascade, even though futures fall further in magnitude at each minute. Tick-level data would be required to detect lead-lag dynamics below 60 seconds. Third, this paper observes two exchanges (Binance and Bybit); data from OKX, Coinbase, and decentralized venues such as Hyperliquid would further strengthen generalizability. Fourth, the cascade model is conceptual and not calibrated to order book depth data; calibration would require depth snapshots that are not retained in standard exchange APIs. Fifth, the geopolitical trigger is documented from secondary sources and the exact timestamp of the originating Truth Social post is not independently verified to sub-second precision; the contribution does not depend on a precise causal identification of the trigger. Sixth, the \$19 billion liquidation figure widely reported in industry sources is a CoinGlass platform-aggregated estimate of notional leverage that is not independently audited and may be revised in subsequent CoinGlass snapshots; this paper does not condition any quantitative finding on this number. The 62-event distributional analysis in Section 6.2 addresses the concern that small-sample comparison could reflect favorable event selection; the statistical position of October 10 within the empirical distribution provides confidence in robustness.

11. Conclusion

Using minute-level data from Binance and Bybit, this paper situates the October 10, 2025 cryptocurrency crash within an empirical distribution of 62 BTC drawdowns exceeding 3% during 2024–2026. October 10 ranks first of 62 on eight of eleven microstructure metrics, including the four most diagnostic of liquidation-cascade behavior: SOL futures-spot gap ($z = +7.65$), BTC basis swing ($z = +6.18$), mark-to-spot undershoot ($z = -4.21$), and BTC volume surge ratio ($z = +4.17$). Multivariate anomaly measures rank it first as well: its Mahalanobis distance of 7.70 exceeds the second-place event by 20 percent, and its sum-of-absolute-z-scores of 45.9 exceeds the second-place event by 142 percent. A parsimonious conceptual cascade model identifies the mark price oracle as the crypto-specific amplifier: an originally protective design feature that, through reflexive feedback into the liquidation engine, plausibly served as the cascade's primary mechanism. The findings are consistent with a structural fragility in cryptocurrency exchange design — endogenous crash amplification through oracle reflexivity — that is absent from traditional market architectures. The evidence suggests that the lessons from the 2010 equity Flash Crash, particularly circuit breakers and margin call windows, remain unlearned in cryptocurrency market design.

Declarations

Use of AI Technology. The author used AI-assisted tools (large language model-based assistants) during the preparation of this manuscript for the following purposes: assisting with data collection scripts written in Python for the Binance and Bybit REST APIs, assisting with statistical computation and verification, and providing editorial suggestions on clarity and structure. All analytical decisions, interpretation of results, and final wording are the author's responsibility. AI tools were not used to generate the substantive scholarly content, claims, or conclusions of the paper.

Conflicts of Interest. The author declares no conflicts of interest.

Informed Consent. This research did not involve human participants. Informed consent is therefore not applicable.

Data Availability. All data are publicly available from Binance and Bybit public REST APIs (Binance spot: <https://api.binance.com/api/v3/klines>; Binance perpetual futures: <https://fapi.binance.com/fapi/v1/klines>; Binance mark price: <https://fapi.binance.com/fapi/v1/markPriceKlines>; Bybit v5: <https://api.bybit.com/v5/market/kline>). The full 28-month BTC and SOL 1-minute kline dataset used to

construct the 62-event distribution in Section 6.2 can be reconstructed from these endpoints. Replication scripts implementing the event-construction algorithm in Appendix A, the cross-event statistics in Section 6, and the multivariate anomaly analysis in Section 6.3 are available at <https://github.com/boonchuan/delta-exchange>.

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Appendix A. Event Construction Algorithm

This appendix specifies the procedure used to construct the 62-event empirical distribution in Section 6.2. The algorithm is deterministic and reproducible from public Binance REST API data.

Step 1. Data acquisition. For BTC and SOL, pull 1-minute klines for spot and perpetual futures contracts (symbols BTCUSDT, SOLUSDT) from the Binance public REST API for the period January 1, 2024 00:00 UTC through April 30, 2026 23:59 UTC. Spot endpoint: <https://api.binance.com/api/v3/klines>. Perpetual futures endpoint: <https://fapi.binance.com/fapi/v1/klines>. For BTC, additionally pull mark price klines from <https://fapi.binance.com/fapi/v1/markPriceKlines>. Pagination is performed using the `startTime` and `endTime` parameters with `limit=1000` candles per request. Duplicate minute candles (rare boundary artifacts at month transitions) are removed by keeping the first occurrence.

Step 2. Drawdown scan. Let $C(t)$ denote the BTC spot close price at minute t . For each minute t with at least 30 prior minutes of data, compute the rolling 30-minute peak-to-trough drawdown as $DD(t) = \min(C(t-30:t)) / \max(C(t-30:t)) - 1$, expressed as a percentage. Retain all minutes for which $DD(t) \leq -3.0\%$. This step typically yields several hundred raw candidate minutes per month during high-volatility periods.

Step 3. De-overlapping by greedy selection. Sort the candidate minutes by $DD(t)$ ascending (most negative first). Initialize an empty list of selected events. For each candidate t in sorted order: if there is no previously selected event whose window-end timestamp lies within 6 hours of t , add t to the selected list with window endpoints $(t-30, t)$; otherwise skip t . The 6-hour separation rule prevents counting multi-stage cascades as multiple events while preserving genuinely distinct shocks separated by at least a quarter-day. Across 28 months this procedure yields 62 events.

Step 4. Crash-window relocation. For each selected event, identify the BTC spot trough minute t^* within the 30-minute window, and define the analysis window as $[t^* - 10, t^* + 20]$. This centered window is used for all per-event microstructure metric computations (basis swing, futures drawdown, volume surge ratio, mark undershoot, intra-minute spread, SOL gap). The 24-hour period immediately preceding the analysis window provides the volume baseline.

Step 5. Metric computation. Within each event's analysis window, compute: (i) BTC spot peak-to-trough drawdown; (ii) BTC futures peak-to-trough drawdown; (iii) basis swing as $\max(F_t - S_t) - \min(F_t - S_t)$ over the window; (iv) mark-to-spot undershoot as the difference between the BTC mark price low and the BTC spot close at the same minute as the mark low; (v) maximum intra-minute spread as $\max((H_t - L_t) / L_t \times 100)$ over the window using BTC spot high and low; (vi) volume surge ratio as the window's mean BTC spot volume divided by the 24-hour pre-window baseline mean; (vii) peak volume ratio as the window's maximum minute volume divided by the same baseline; (viii) volume lead minutes as the time difference (in minutes) between the BTC spot price trough minute and the BTC spot volume peak minute; (ix) SOL spot drawdown, SOL futures drawdown, and SOL futures-spot gap, computed analogously.

Step 6. Sign conventions and z-scoring. Drawdown metrics (BTC spot DD, BTC futures DD, SOL spot DD, SOL futures DD, mark-to-spot undershoot) are sign-flipped to convert "more extreme = more negative" into "more extreme = more positive" for purposes of computing the sum-of-absolute-z-scores composite anomaly. Z-scores are computed against the empirical mean and standard deviation of each metric within the 62-event sample. Mahalanobis distance is computed in z-score space using the sample covariance matrix and its inverse (or pseudo-inverse if numerically singular).

Replication scripts implementing this algorithm in Python (using the requests, pandas, numpy, and scipy libraries) will be made available upon publication. Missing minute candles (rare boundary artifacts) are handled by forward-fill of the immediately preceding candle, which does not materially affect any metric reported in this paper because the only events with missing minutes are not among the detected drawdowns.

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